Kidzee Polichalur — Requirements Gap Analysis

> ***Version:*** *1.0 (Pre-questionnaire)* ***Date:*** *March 29, 2026* ***Status:*** *Pending stakeholder input*

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1. Current State vs Requirements

#	Requirement (from Imtiaz)	Current Status	Gap	Priority
1	Centralized activity platform	**DONE** — Home page, activity listing, year browsing	None	-
2	Public access for everyone	**DONE** — No auth required for public pages	None	-
3	Admin access for year-wise uploads	**DONE** — Password-protected admin CRUD	None	-
4	Auto-publish to Google	**GAP** — No Google Business Profile integration	Build or integrate	High
5	Auto-publish to YouTube	**GAP** — Video URLs supported but no upload	Build or integrate	High
6	Auto-publish to Instagram	**GAP** — No Instagram API integration	Build or integrate	High
7	Auto-publish to Facebook	**GAP** — No Facebook API integration	Build or integrate	High
8	"Similar platforms for visibility"	**PARTIAL** — Website exists, but no SEO/OG tags	Enhance	Medium

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2. Phased Implementation Roadmap

Phase 1: Quick Wins (Week 1)

No API keys needed, no stakeholder setup required

- [ ] Add Open Graph meta tags to all activity pages (so links look good when shared on WhatsApp/Facebook)
- [ ] Add "Share" buttons per activity (WhatsApp, Facebook, Instagram, Copy Link)
- [ ] Add SEO meta tags + sitemap.xml for Google indexing
- [ ] Add school branding (logo, colors, name from questionnaire)
- [ ] Activity image gallery with lightbox

Phase 2: Social Media Integration (Week 2-3)

Requires stakeholder social media accounts + API access

- [ ] Facebook Page auto-posting via Meta Graph API
- [ ] Instagram auto-posting via Instagram Content Publishing API

- [ ] YouTube video upload via YouTube Data API v3
- [ ] Google Business Profile posting via GBP API
- [ ] Admin UI: toggle which platforms to post to per activity
- [ ] Admin UI: preview post before publishing
- [ ] Optional: scheduling (post at a specific time)

Phase 3: Communication & Analytics (Week 3-4)

Requires parent contact info + WhatsApp setup

- [ ] WhatsApp sharing to parent groups (via WhatsApp Business API or share links)
- [ ] Activity view counter / basic analytics
- [ ] Monthly activity digest (email or WhatsApp)
- [ ] Custom domain setup (if requested)
- [ ] Image watermarking with school logo

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3. Social Media Integration — Tool Options

Option A: Direct API Integration (Build It)

Build social media posting directly into the app.

Platform	API	Free Tier	Limitations
Facebook	Meta Graph API	Free	Requires Facebook Page, App Review for public use
Instagram	Instagram Content Publishing API	Free	Business account required, no Stories/Reels via API
YouTube	YouTube Data API v3	10,000 units/day free	Upload = 1,600 units, ~6 uploads/day
Google Business	Google Business Profile API	Free	Requires verified business listing

Pros: Full control, no monthly cost, integrated UX **Cons:** API setup complexity, each platform has approval process, maintenance burden

Option B: Middleware (Buffer / Zapier)

Use a third-party tool to handle social media posting.

Tool	Free Tier	Paid	Features
Buffer	3 channels, 10 scheduled posts	$6/mo/channel	Multi-platform scheduling, analytics
Hootsuite	Limited free	$99/mo	Enterprise features, team management

Zapier	100 tasks/mo	$20/mo (750 tasks)	Webhook trigger → post to any platform
Make.com	1,000 ops/mo	$9/mo	Visual automation builder
IFTTT	2 applets free	$3.50/mo	Simple if-this-then-that automation

Pros: Quick setup, handles API complexity, multi-platform out of the box **Cons:** Monthly cost, dependency on third party, less customizable

Option C: Hybrid (Recommended)

- **Phase 1:** Share buttons + OG tags (zero cost, immediate value)
- **Phase 2:** Zapier/Make.com webhook from our app → auto-post (low cost, fast to build)
- **Phase 3:** Replace with direct API if scale/cost justifies it

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4. Media Storage Options

Option	Free Tier	Cost at Scale	Pros	Cons
Vercel Blob	256 MB free	$0.15/GB/mo	Already on Vercel, simple API	Limited free tier
Cloudinary	25 GB storage, 25 GB bandwidth	Pay as you go	Auto-resize, optimization, transformations	Another service to manage
Supabase Storage	1 GB free	$0.021/GB/mo	Generous free tier, good API	Needs Supabase project
Local JSON (current)	Unlimited (on disk)	Serverless = ephemeral	Zero setup	Data lost on redeploy

Current issue: The app uses file-based JSON storage (`data/activities.json`) which doesn't persist on Vercel serverless. **This needs to be migrated to a database/storage service before going to production.**

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5. Pending — Awaiting Questionnaire Responses

The following decisions are blocked until Imtiaz responds:

1. **Which social media platforms** to prioritize (may not need all 4)
2. **Auto-post vs manual share** preference (determines build complexity)
3. **Budget** — determines whether we use free APIs or middleware
4. **Existing social media accounts** — determines API setup path
5. **Content approval workflow** — auto-publish or review first?
6. **Parent notification preference** — WhatsApp, email, or just website?
7. **Storage migration** — need to decide on Supabase vs Cloudinary vs Vercel Blob
8. **Custom domain** — if yes, need to set up DNS

This document will be updated after receiving the completed questionnaire from Imtiaz.